Characterizing Datasets: Hardness, Learning, and More

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Abstract

We develop elementary Fourier analysis *over a finite dataset* $\mathcal{D} \subset \mathcal{T}^n$: coefficients are taken against the empirical measure on $\mathcal{D}$, while the characters stay those of the ambient product space, so every quantity is estimable from samples. Parseval fails over a dataset, off by exactly the density factor $C_{\mathcal{D}} = |\mathcal{T}|^n \ / \ |\mathcal{D}|$; the replacements are a mass identity, a closeness identity, and a low-degree learning theorem whose normalization constants cancel through a scaled reconstruction. We then adapt the Goldreich-Levin algorithm to find every character heavily correlated with a function *over the dataset*. Sample access alone is blind: over a collision-free dataset, every bucket weight in the classical tree search is identical. Query access to a model, together with the dataset's heavy bias spectrum, suffices — via a convolution identity writing the on-dataset spectrum as the global spectrum convolved with the dataset's own spectrum. Corollaries include a Kushilevitz-Mansour guarantee for decision trees over datasets.

1 Fourier Analysis on a Dataset

Classical Fourier analysis of Boolean functions — and of functions on product probability spaces generally — takes expectations over a *product measure*: every coordinate is resampled independently. Real data is not like this. A corpus of token sequences, a folder of images, a table of feature vectors — each is a finite set $\mathcal{D} \subset \mathcal{T}^n$ inside an ambient product space, and the functions we care about (a trained model, a labeling rule) are only evaluated *on the data*.

This note keeps the characters of the ambient product space but takes all expectations over the dataset. The resulting "dataset Fourier coefficients" are easy to estimate from samples and keep much of the classical dictionary — but not all of it, and the failures are precise, quantifiable, and (in the Goldreich-Levin section) algorithmically useful. Throughout, f is a trained model, an ideal labeling function on the dataset, or another efficiently computable function.

1.1 Orthonormal Bases and Parseval

The ambient framework is standard [31].

Definition 1.1 *(Probability Space with Orthonormal Basis)*: Let Ω be a measurable space with probability measure μ, and let I be an index set. An *orthonormal basis* for $L^2(\Omega, \mu)$ is a collection $\{\varphi_\alpha\}_{\alpha \in I}$ of functions $\varphi_\alpha : \Omega \to \mathbb{R}$ satisfying:

1. **Orthonormality:** $\langle \varphi_\alpha, \varphi_\beta \rangle_\mu = \mathbb{E}_{x \sim \mu}[\varphi_\alpha(x) \varphi_\beta(x)] = \delta_{\alpha\beta}$;
2. **Completeness:** every $f \in L^2(\Omega, \mu)$ can be written as $f = \sum_{\alpha \in I} \hat{f}(\alpha) \varphi_\alpha$.

We always take $\varphi_0 = 1$. For product spaces Ω^n with product measure $\mu^{\otimes n}$, the products $\varphi_\alpha = \prod_i \varphi_{\alpha_i}$ (over multi-indices α) form an orthonormal basis for $L^2(\Omega^n, \mu^{\otimes n})$.

Definition 1.2 *(Fourier Coefficients)*: For $f \in L^2(\Omega^n, \mu)$, the *Fourier coefficient* at α is

$$\hat{f}(\alpha) = \langle f, \varphi_\alpha \rangle_\mu = \mathbb{E}_{x \sim \mu}[f(x) \varphi_\alpha(x)],$$

so that $f = \sum_\alpha \hat{f}(\alpha) \varphi_\alpha$ (in L^2).

Proposition 1.1 *(Parseval)*: For any $f, g \in L^2(\Omega^n, \mu)$,

$$\langle f, g \rangle_\mu = \sum_\alpha \hat{f}(\alpha) \hat{g}(\alpha), \quad \text{in particular} \quad \mathbb{E}_{x \sim \mu}[f(x)^2] = \sum_\alpha \hat{f}(\alpha)^2.$$

Proof: Expand both functions in the basis and use orthonormality; the interchange of sum and integral is justified by L^2 convergence. ■

Two running examples:

Example: *(Haar / parity basis.)* $\Omega = \{-1, 1\}$ with the uniform measure. For $S \subseteq [n]$ the *parity functions* $\chi_S(x) = \prod_{i \in S} x_i$ form an orthonormal basis of $L^2(\{-1, 1\}^n)$: for $S \neq T$ pick $i \in S \Delta T$ and note $\mathbb{E}[\chi_{S \Delta T}] = \mathbb{E}[x_i] \cdot \mathbb{E}\left[\chi_{S \Delta T \setminus \{i\}}\right] = 0$.

Example: *(Hermite basis.)* $\Omega = \mathbb{R}$ with the standard Gaussian measure γ. The orthonormalized *probabilist's Hermite polynomials* $h_0 = 1$, $h_1(x) = x$, $h_2(x) = (x^2 - 1)/\sqrt{2}, ...$ form an orthonormal basis; their products $H_\alpha(x) = \prod_i h_{\alpha_i}(x_i)$, $\alpha \in \mathbb{N}^n$, handle Gaussian space $L^2(\mathbb{R}^n, \gamma_n)$, with $\deg H_\alpha = |\alpha| = \sum_i \alpha_i$. (The Ornstein-Uhlenbeck noise operator and the invariance principle port over as well; we will not need them here.)

1.2 Coordinate Averaging and Sensitivity

A standard way to measure how much f depends on a coordinate:

Definition 1.3 *(Coordinate Averaging and Sensitivity)*: For $i \in [n]$, the *averaging operator* is $E^i[f](x) = \mathbb{E}_{a \sim \mu_i}[f(x^{i \leftarrow a})]$, where $x^{i \leftarrow a}$ replaces the i-th coordinate of x by a. The *sensitivity of coordinate i* (often called *influence*) and the *total sensitivity* are

$$\mathrm{Sens}_i[f] = \mathbb{E}_{x \sim \mu}\left[\left(f(x) - E^i[f](x)\right)^2\right], \qquad \boldsymbol{S}[f] = \sum_{i=1}^n \mathrm{Sens}_i[f].$$

Proposition 1.2: E^i kills exactly the basis directions that use coordinate i: $E^i[\varphi_\alpha] = \varphi_\alpha$ if $\alpha_i = 0$ and $E^i[\varphi_\alpha] = 0$ otherwise. Consequently $\mathrm{Sens}_i[f] = \sum_{\alpha : \alpha_i \neq 0} \hat{f}(\alpha)^2$ and $\boldsymbol{S}[f] = \sum_\alpha \#\alpha \cdot \hat{f}(\alpha)^2$, where $\#\alpha = |\{i : \alpha_i \neq 0\}|$.

Proof: The first claim is orthonormality in the i-th factor ($\mathbb{E}_{a \sim \mu_i}\left[\varphi_{\alpha_i}(a)\right] = \left\langle \varphi_{\alpha_i}, \varphi_0 \right\rangle = \delta_{\alpha_i, 0}$); the rest is Parseval applied to $f - E^i f = \sum_{\alpha : \alpha_i \neq 0} \hat{f}(\alpha) \varphi_\alpha$. ■

This is the spectral form of the total-effect (Sobol) index from variance-based global sensitivity analysis [26, 35]; estimated over data, it underlies variance-based feature attribution [12] — a connection the dataset version below makes literal.

1.3 Datasets: the Basic Dictionary

Now the object of study. Fix a finite alphabet $\mathcal{T}$ and — from here to the end — let μ be the **uniform** measure on $\mathcal{T}$, with $\{\varphi_\alpha\}$ any orthonormal basis of $L^2(\mathcal{T}^n)$ containing $\varphi_0 = 1$. (We note below what survives for non-uniform μ.)

Definition 1.4 *(Dataset; Dataset Fourier Coefficient)*: A *dataset* is a finite nonempty set $\mathcal{D} \subset \mathcal{T}^n$; we write $x \sim \mathcal{D}$ for a uniformly random element of $\mathcal{D}$ and

$$\hat{f}_{\mathcal{D}}(\alpha) = \mathbb{E}_{x\sim\mathcal{D}}[f(x)\varphi_\alpha(x)] = \frac{1}{|\mathcal{D}|}\sum_{x\in\mathcal{D}} f(x)\varphi_\alpha(x)$$

for the *dataset Fourier coefficient* of $f : \mathcal{D} \to \mathbb{R}$ at α. The *density constant* is $C_{\mathcal{D}} = |\mathcal{T}|^n \,/\, |\mathcal{D}|$.

Dataset coefficients are exactly the quantities estimable from labeled samples $(x, f(x))$, $x \sim \mathcal{D}$ — that is the point of the definition. The catch: $\{\varphi_\alpha\}$ is orthonormal in $L^2(\mathcal{T}^n, \mu)$, *not* in $L^2(\mathcal{D})$. Two characters may even coincide as functions on $\mathcal{D}$ (we call this *aliasing*; the subcube example in the next section shows it). So the Parseval-based dictionary cannot be borrowed blindly. The replacement dictionary routes through one object:

Definition 1.5 *(The Lift)*: For $f : \mathcal{D} \to \mathbb{R}$, the *lift* $\mathcal{D} \circ f : \mathcal{T}^n \to \mathbb{R}$ is

$$(\mathcal{D} \circ f)(x) = \begin{cases} f(x) & \text{if } x \in \mathcal{D} \\ 0 & \text{otherwise.} \end{cases}$$

Lemma 1.1 *(Lifting Lemma)*: $(\widehat{\mathcal{D} \circ f})(\alpha) = C_{\mathcal{D}}^{-1}\hat{f}_{\mathcal{D}}(\alpha)$ for every α.

Proof: $\mathbb{E}_{x\sim\mu}[(\mathcal{D}\circ f)(x)\varphi_\alpha(x)] = \frac{1}{|\mathcal{T}|^n}\sum_{x\in\mathcal{D}} f(x)\varphi_\alpha(x) = \frac{|\mathcal{D}|}{|\mathcal{T}|^n}\cdot\hat{f}_{\mathcal{D}}(\alpha)$. ■

Every dataset fact in this note is the Lifting Lemma composed with a fact about $L^2(\mu)$. Three instances follow. First, inversion — the dataset coefficients determine f on $\mathcal{D}$:

Proposition 1.3 *(Inversion)*: For every $x \in \mathcal{D}$: $\quad f(x) = C_{\mathcal{D}}^{-1}\sum_\alpha \hat{f}_{\mathcal{D}}(\alpha)\varphi_\alpha(x)$.

Proof: Expand the lift in $L^2(\mu)$ and evaluate at $x \in \mathcal{D}$, where $(\mathcal{D}\circ f)(x) = f(x)$; apply Lemma 1.1. ■

Second, the replacement for Parseval: there is no Parseval identity over $\mathcal{D}$, and the truth is off by exactly the density constant.

Theorem 1.1 (Mass Identity — "Parseval over a dataset"): For any dataset $\mathcal{D}$ (distinct points) and any $f : \mathcal{D} \to \mathbb{R}$,

$$\sum_\alpha \hat{f}_{\mathcal{D}}(\alpha)^2 = C_{\mathcal{D}} \cdot \mathbb{E}_{x\sim\mathcal{D}}[f(x)^2].$$

Proof: Since $\{\varphi_\alpha\}$ is a complete orthonormal system for the uniform measure on the finite set $\mathcal{T}^n$, it satisfies the *reproducing kernel identity*

$$\sum_\alpha \varphi_\alpha(x)\varphi_\alpha(x') = |\mathcal{T}|^n \cdot \mathbb{1}[x = x']$$

(the matrix $M_{\alpha,x} = \varphi_\alpha(x)/\sqrt{|\mathcal{T}|^n}$ is square with orthonormal rows, hence orthonormal columns[1]). Now expand and swap sums:

$$\sum_\alpha \hat{f}_\mathcal{D}(\alpha)^2 = \frac{1}{|\mathcal{D}|^2} \sum_{x,x' \in \mathcal{D}} f(x)f(x') \sum_\alpha \varphi_\alpha(x)\varphi_\alpha(x') = \frac{|\mathcal{T}|^n}{|\mathcal{D}|^2} \sum_{x \in \mathcal{D}} f(x)^2 = C_\mathcal{D} \cdot \mathbb{E}_\mathcal{D}[f^2].$$

■

The $1/\ |\mathcal{D}|^2$ prefactor of the double sum might suggest $C_\mathcal{D}^2$, but the kernel collapses the double sum to its diagonal, and one factor of $1/\ |\mathcal{D}|$ is absorbed into $\mathbb{E}_\mathcal{D}[f^2] = \frac{1}{|\mathcal{D}|} \sum_{x \in \mathcal{D}} f(x)^2$ — so exactly one factor of $C_\mathcal{D} = |\mathcal{T}|^n\ /\ |\mathcal{D}|$ survives, not its square. (The appendix's numerical checks confirm the ratio $\sum_\alpha \hat{f}_\mathcal{D}(\alpha)^2 / \mathbb{E}_\mathcal{D}[f^2]$ equals $C_\mathcal{D}$ exactly.)

For a non-uniform finite base measure with full support the same proof gives the weighted form $\sum_\alpha \hat{f}_\mathcal{D}(\alpha)^2 = \frac{1}{|\mathcal{D}|^2} \sum_{x \in \mathcal{D}} f(x)^2 / \mu(x)$; for continuous Ω (e.g., Gaussian space) the left side diverges — the empirical measure of a finite dataset has no $L^2(\mu)$ density. This is why we fix a finite alphabet.

Since $C_\mathcal{D}$ is large for small datasets, the total dataset Fourier mass is *not* $\mathbb{E}_\mathcal{D}[f^2]$: a function of unit norm on the dataset carries $C_\mathcal{D}$ worth of squared coefficients, smeared across aliases. The constant is a normalization, not a defect. Dividing each restricted character by $\sqrt{C_\mathcal{D}}$ gives a normalized system in which every identity holds over the dataset with no constants.

Definition 1.6 *(Normalized Dataset Coefficients)*: The *normalized characters* and *normalized dataset coefficients* are

$$\breve{\varphi}_\alpha = C_\mathcal{D}^{-1/2} \varphi_\alpha, \qquad \breve{f}_\mathcal{D}(\alpha) = \langle f, \breve{\varphi}_\alpha \rangle_\mathcal{D} = C_\mathcal{D}^{-1/2} \hat{f}_\mathcal{D}(\alpha).$$

Theorem 1.2 (Parseval and Closeness over the Dataset): For all $f, g : \mathcal{D} \to \mathbb{R}$:
1. (Parseval) $\sum_\alpha \breve{f}_\mathcal{D}(\alpha)^2 = \mathbb{E}_{x \sim \mathcal{D}}[f(x)^2]$;
2. (Closeness) $\mathbb{E}_{x \sim \mathcal{D}}[(f(x) - g(x))^2] = \sum_\alpha \left(\breve{f}_\mathcal{D}(\alpha) - \breve{g}_\mathcal{D}(\alpha)\right)^2$;
3. (Reconstruction) $f(x) = \sum_\alpha \breve{f}_\mathcal{D}(\alpha) \breve{\varphi}_\alpha(x)$ for every $x \in \mathcal{D}$.

Proof: (1) is Theorem 1.1 divided by $C_\mathcal{D}$; (2) is (1) applied to $f - g$, using linearity of $\mathbb{E}_\mathcal{D}$; (3) is Proposition 1.3, with the two factors of $C_\mathcal{D}^{-1/2}$ recombining into $C_\mathcal{D}^{-1}$. ■

Both sides of Theorem 1.2 are over the dataset — no expectation over $\mathcal{T}^n$ appears — with no stray constants. The normalization just bookkeeps the overcompleteness: $L^2(\mathcal{D})$ has dimension $|\mathcal{D}|$, yet there are $|\mathcal{T}|^n$ restricted characters, whose squared norms over $\mathcal{D}$ *average* exactly $C_\mathcal{D}^{-1}$ by the kernel diagonal $\sum_\alpha \varphi_\alpha(x)^2 = |\mathcal{T}|^n$; aliasing is this overcompleteness made visible. The raw coefficient $\hat{f}_\mathcal{D}(\alpha) = \mathbb{E}_\mathcal{D}[f \varphi_\alpha]$ is the *estimable correlation* — what a sampling algorithm sees, and the unit for heavy-coefficient search in the Goldreich-Levin section — while $\breve{f}_\mathcal{D}(\alpha)$ is the unit for energy and closeness; converting between them costs one factor of $\sqrt{C_\mathcal{D}}$.

Definition 1.7 *($\epsilon_\mathcal{D}$-closeness; Normalized Weights)*: We say g is *$\epsilon_\mathcal{D}$-close* to f if $\mathbb{E}_{x \sim \mathcal{D}}[(f(x) - g(x))^2] \le \epsilon$. The *normalized level weights* are

$$\overline{W}_\mathcal{D}^k[f] = \sum_{\#\alpha = k} \breve{f}_\mathcal{D}(\alpha)^2, \quad \text{so that} \quad \sum_{k \ge 0} \overline{W}_\mathcal{D}^k[f] = \mathbb{E}_\mathcal{D}[f^2]$$

by Theorem 1.2.

[1]Squareness: since the uniform measure has full support, orthonormality forces $|I| \le |\mathcal{T}|^n$, and completeness forces $|I| = |\mathcal{T}|^n$.

1.4 Averaging and Sensitivity on a Dataset

The third instance of the lift-then-use-$L^2(\mu)$ recipe is coordinate averaging. On a dataset, resampling a coordinate may leave the data; the natural operator zeros out such settings, which makes it agree exactly with the global operator applied to the lift.

Definition 1.8 *(Dataset Coordinate Averaging; Dataset Sensitivity)*: For $x \in \mathcal{D}$,

$$E^i_{\mathcal{D}}[f](x) = \mathbb{E}_{a\sim\mu_i}[\mathcal{D}(x^{i\leftarrow a}) \cdot f(x^{i\leftarrow a})], \qquad \mathrm{Sens}^i_{\mathcal{D}}[f] = \mathbb{E}_{x\sim\mathcal{D}}\left[\left(f(x) - E^i_{\mathcal{D}}[f](x)\right)^2\right],$$

where $\mathcal{D}(\cdot)$ denotes the indicator of the dataset. Note $E^i_{\mathcal{D}}[f] = E^i[\mathcal{D} \circ f]$ restricted to $\mathcal{D}$, directly from the definitions.

Proposition 1.4 *(Dataset Sensitivity Bound)*:

$$\mathrm{Sens}^i_{\mathcal{D}}[f] \le C_{\mathcal{D}} \cdot \mathbb{E}_{x\sim\mu}\left[\left((\mathcal{D} \circ f) - E^i[\mathcal{D} \circ f]\right)^2\right] = \sum_{\alpha:\alpha_i\neq 0} \breve{f}_{\mathcal{D}}(\alpha)^2,$$

and the inequality is an equality when $(\mathcal{D} \circ f) - E^i[\mathcal{D} \circ f]$ is supported on $\mathcal{D}$. Consequently $\boldsymbol{S}_{\mathcal{D}}[f] \coloneqq \sum_i \mathrm{Sens}^i_{\mathcal{D}}[f] \le \sum_{k\ge 1} k \cdot \overline{W}^k_{\mathcal{D}}[f]$.

Proof: Write $q = (\mathcal{D} \circ f) - E^i[\mathcal{D} \circ f]$, so that $\mathrm{Sens}^i_{\mathcal{D}}[f] = \mathbb{E}_{\mathcal{D}}[q^2] = C_{\mathcal{D}} \cdot \mathbb{E}_{\mu}[\mathcal{D} \cdot q^2] \le C_{\mathcal{D}} \cdot \mathbb{E}_{\mu}[q^2]$. Then apply Proposition 1.2 to the lift and Lemma 1.1; summing over i counts each α once per nonzero coordinate. ∎

The inequality can be strict, and the naive analogue of Proposition 1.2 with dataset coefficients (no $C_{\mathcal{D}}^{-1}$) is false. Example: $\mathcal{D} = \{x : x_1 = 1\} \subsetneq \{-1,1\}^n$ and $f = x_2$. Then $\mathrm{Sens}^2_{\mathcal{D}}[f] = 1$, while $\chi_{\{2\}}$ and $\chi_{\{1,2\}}$ alias on $\mathcal{D}$, so $\sum_{\alpha:\alpha_2\neq 0} \hat{f}_{\mathcal{D}}(\alpha)^2 = 1 + 1 = 2$; in normalized units the bound of Proposition 1.4 reads $\sum_{\alpha:\alpha_2\neq 0} \breve{f}_{\mathcal{D}}(\alpha)^2 = 2/C_{\mathcal{D}} = 1$ (here $C_{\mathcal{D}} = 2$) — tight.

1.5 Learning Low-Degree Functions from Dataset Samples

The classical Low-Degree Algorithm [31] survives on datasets, *provided* the reconstruction is the truncated *normalized* expansion $\sum_{\#\alpha\le d} \breve{f}_{\mathcal{D}}(\alpha)\breve{\varphi}_\alpha$ — equivalently, the raw plug-in reconstruction scaled by $C_{\mathcal{D}}^{-1}$. The scaling matters. In the half-cube example above, the unscaled plug-in reconstruction of $f = x_2$ at degree 2 returns $\chi_2 + \chi_{12} = 2x_2$ on $\mathcal{D}$ — error 1, despite exact coefficients and zero tail — because each character is counted once per alias. The scaled reconstruction returns $(x_2 + x_1x_2)/2 = x_2$ on $\mathcal{D}$ exactly, by part (3) of Theorem 1.2 and the fact that all coefficients outside $\{\chi_2, \chi_{12}\}$ vanish there.

Two conventions for the statement. The one-coordinate basis is *B-bounded*: $\|\varphi_j\|_\infty \le B$ for all j, so $\|\varphi_\alpha\|_\infty \le B^d$ whenever $\#\alpha \le d$; for the parity characters, and any group-character basis, $B = 1$. Write $N_d = |\{\alpha : \#\alpha \le d\}| = \sum_{k\le d}\binom{n}{k}(|\mathcal{T}| - 1)^k \le (1 + n(|\mathcal{T}| - 1))^d$ for the number of low-degree indices.

Theorem 1.3 (Learning Low-Degree Functions over a Dataset): Let $f : \mathcal{D} \to [-1, 1]$ and suppose the normalized high-degree weight satisfies

$$\sum_{k>d} \overline{W}^k_{\mathcal{D}}[f] \le \frac{\epsilon}{4}$$

(e.g., $d \ge 4\overline{S}/\epsilon$ suffices when $\sum_k k \cdot \overline{W}^k_{\mathcal{D}}[f] \le \overline{S}$). Given $|\mathcal{D}|$ and $m = O\Big(B^{2d} \frac{N_d}{\epsilon} \cdot \log(N_d/\delta)\Big)$ samples from $\mathcal{D}$, one can output h that is $\epsilon_{\mathcal{D}}$-close to f with probability at least $1 - \delta$.

Proof: **Algorithm.** For each $\#\alpha \le d$, estimate $\hat{f}_{\mathcal{D}}(\alpha)$ by $\tilde{f}(\alpha) = \frac{1}{m} \sum_{j=1}^m f\big(x^{(j)}\big)\varphi_\alpha\big(x^{(j)}\big)$; output the *scaled* reconstruction $h = C_{\mathcal{D}}^{-1} \sum_{\#\alpha \le d} \tilde{f}(\alpha)\varphi_\alpha$ (this scaling is where knowledge of $|\mathcal{D}|$ enters).

Analysis. Write $g^\uparrow = \mathcal{D} \circ f$, so $\widehat{g^\uparrow}(\alpha) = C_{\mathcal{D}}^{-1} \hat{f}_{\mathcal{D}}(\alpha)$ (Lemma 1.1) and $f = g^\uparrow$ on $\mathcal{D}$. Dropping the indicator and applying Parseval in $L^2(\mu)$:

$$\begin{aligned}\mathbb{E}_{\mathcal{D}}[(f-h)^2] = C_{\mathcal{D}} \cdot \mathbb{E}_\mu\Big[\mathcal{D} \cdot (g^\uparrow - h)^2\Big] &\le C_{\mathcal{D}} \cdot \mathbb{E}_\mu\Big[(g^\uparrow - h)^2\Big] \\ &= C_{\mathcal{D}} \sum_\alpha \Big(\widehat{g^\uparrow}(\alpha) - \hat{h}(\alpha)\Big)^2.\end{aligned}$$

Now $\hat{h}(\alpha) = C_{\mathcal{D}}^{-1} \tilde{f}(\alpha)$ for $\#\alpha \le d$ and 0 otherwise, while $\widehat{g^\uparrow}(\alpha) = C_{\mathcal{D}}^{-1} \hat{f}_{\mathcal{D}}(\alpha)$ throughout, so each squared difference carries a factor ${C_{\mathcal{D}}^{-1}}^2$; since $C_{\mathcal{D}} \cdot {C_{\mathcal{D}}^{-1}}^2 = C_{\mathcal{D}}^{-1}$, splitting the sum at degree d gives

$$\mathbb{E}_{\mathcal{D}}[(f-h)^2] \le \underbrace{C_{\mathcal{D}}^{-1} \sum_{\#\alpha > d} \hat{f}_{\mathcal{D}}(\alpha)^2}_{= \sum_{k>d} \overline{W}^k_{\mathcal{D}}[f]\ ;\ \le \epsilon/4 \text{ (hyp.)}} + \underbrace{C_{\mathcal{D}}^{-1} \sum_{\#\alpha \le d} \Big(\hat{f}_{\mathcal{D}}(\alpha) - \tilde{f}(\alpha)\Big)^2}_{\le \epsilon/4 \text{ w.h.p.}},$$

where the first bracket used $C_{\mathcal{D}}^{-1} \hat{f}_{\mathcal{D}}(\alpha)^2 = \breve{f}_{\mathcal{D}}(\alpha)^2$ (Definition 1.6) and $\overline{W}^k_{\mathcal{D}}[f] = \sum_{\#\alpha = k} \breve{f}_{\mathcal{D}}(\alpha)^2$. For the second term: each $\tilde{f}(\alpha)$ averages m i.i.d. terms bounded by $\|f\|_\infty \|\varphi_\alpha\|_\infty \le B^d$ with mean $\hat{f}_{\mathcal{D}}(\alpha)$; by Hoeffding (range $2B^d$) and a union bound over the N_d low-degree indices, the stated m makes every deviation at most $\frac{1}{2}\sqrt{\epsilon/N_d}$, so the sum is at most $N_d \cdot \epsilon/(4N_d) = \epsilon/4$ even before the (generous) factor $C_{\mathcal{D}}^{-1} \le 1$. ■

The hypothesis is stated in normalized weights, the analogue of "ϵ of Fourier weight above degree d"; in raw dataset units it reads $\sum_{k>d} \sum_{\#\alpha=k} \hat{f}_{\mathcal{D}}(\alpha)^2 \le C_{\mathcal{D}}\epsilon/4$. For a *generic* sparse dataset it is demanding — it asks the lift $\mathcal{D} \circ f$ to be low-degree concentrated in $L^2(\mu)$ — and the next section shows this is a structural fact about datasets, not an artifact of the proof. Structured datasets (dense ones, or subcube-like ones as above) satisfy it naturally.

1.6 Notation Summary

The recurring symbols (the last four families are introduced in the Goldreich-Levin section):

Symbol	**Meaning**
$\hat{f}(\alpha)$	global Fourier coefficient $\mathbb{E}_{x\sim\mu}[f\varphi_\alpha]$
$\hat{f}_{\mathcal{D}}(\alpha)$	dataset coefficient $\mathbb{E}_{x\sim\mathcal{D}}[f\varphi_\alpha]$ — the estimable correlation
$\breve{f}_{\mathcal{D}}(\alpha) = C_{\mathcal{D}}^{-1/2} \hat{f}_{\mathcal{D}}(\alpha)$	normalized coefficient — the unit for energy and closeness
$C_{\mathcal{D}} = \lvert\mathcal{T}\rvert^n \ /\ \lvert\mathcal{D}\rvert$	density constant; the exact factor in the mass identity

Symbol	Meaning
$\mathcal{D} \circ f$	the lift: f on $\mathcal{D}$, 0 elsewhere; $\widehat{(\mathcal{D} \circ f)} = C_{\mathcal{D}}^{-1} \hat{f}_{\mathcal{D}}$
$\overline{W}_{\mathcal{D}}^{k}[f] = \sum_{\#\alpha=k} \breve{f}_{\mathcal{D}}(\alpha)^2$	normalized level weight; $\sum_k \overline{W}^k = \mathbb{E}_{\mathcal{D}}[f^2]$
b_V, B_θ	bias spectrum of $\mathcal{D}$ (dataset coefficients of 1); its θ-heavy set
$\overline{v}_S(z)$, $\Psi(S\vert J)$	average with context z fixed; conditional bucket weight $\mathbb{E}_z[\overline{v}_S(z)^2]$
c_k, R_k	context profile (distinct contexts at level k); level mass $\sum_S \Psi$
N_k, N	live-bucket profile: buckets with $\Psi \geq \tau^2/4$ at level k; its maximum

2 Goldreich-Levin over a Dataset

We adapt the Goldreich-Levin algorithm [18] ([31], Section 3.5) to the dataset setting. Given query access to f, the classical algorithm finds *all* characters with which f is noticeably correlated; it is what powers the Kushilevitz-Mansour learning algorithm [29]. We want the on-distribution version: f is a trained model (or labeling rule), $\mathcal{D}$ is the data, and we want every character noticeably correlated with f *over the dataset*, i.e. every heavy dataset coefficient $\hat{f}_{\mathcal{D}}(S)$. We also require that the algorithm *never leaves the data*: no membership tester for $\mathcal{D}$, and no evaluation of f on synthetic, out-of-distribution inputs. The replacement resource is one real corpora offer: *conditional sampling* — retrieving datapoints that share a fixed substring (a context), the way an n-gram index retrieves them from a text corpus.

We work over the Boolean cube $\Omega^n = \{-1, 1\}^n$ with the parity characters χ_S, writing $\hat{f}_{\mathcal{D}}(S) = \mathbb{E}_{x\sim\mathcal{D}}[f(x)\chi_S(x)]$ as before.[2] The goal, mirroring the classical guarantee:

> **Definition 2.1** *(Dataset Heavy-Coefficient Search)*: Given $0 < \tau \leq 1$ and access to f and $\mathcal{D}$ (in a model to be specified), output a list L of subsets of $[n]$ such that with high probability:
> 1. (Completeness) $|\hat{f}_{\mathcal{D}}(S)| \geq \tau \implies S \in L$;
> 2. (Soundness) $S \in L \implies |\hat{f}_{\mathcal{D}}(S)| \geq \tau/2$.

2.1 Access Models

The choice of access model is decisive. The underlying object is a corpus $\mathcal{D} \subset \{-1, 1\}^n$ with f on $\mathcal{D}$; the algorithm reaches it through:

1. **Samples** (SAMP): i.i.d. draws $x \sim \mathcal{D}$ together with the label $f(x)$.
2. **Conditional samples** (CSAMP): given coordinates $\overline{J} \subseteq [n]$ and a context z realized in the data, a draw $x \sim \mathcal{D}$ conditioned on $x_{\overline{J}} = z$, together with $f(x)$. This is the "retrieve strings containing this substring" primitive; it is implemented exactly by an index over the corpus (an n-gram or suffix-array structure), or by holding an explicit copy — or large subsample — of $\mathcal{D}$. It is the *subcube-conditioning* oracle studied in distribution testing and learning [9–11], specialized here to the empirical support of $\mathcal{D}$ — and put to a different use: recovering heavy Fourier coefficients of a function f, rather than testing a distribution.

[2] Every identity below that uses only orthonormality and the reproducing kernel $\sum_S \chi_S(x)\chi_S(x') = 2^n \cdot \mathbb{1}[x = x']$ extends verbatim to any finite abelian group with its character basis (with $\chi_S\overline{\chi_T}$ in place of $\chi_S\chi_T$), and to general product orthonormal bases over finite sets. The convolution identity additionally uses the group structure $\chi_S\chi_T = \chi_{S\Delta T}$.

That is the entire model. Absent: any membership tester for $\mathcal{D}$, any evaluation of f outside the data, any synthetic input at all — every string the algorithm touches is a datapoint. (The companion result Theorem 2.3 uses the strictly stronger setting where f is also a cheaply queryable model; we keep that out of the main theorem to show it is not needed.)

Classical GL is of the stronger type — it queries f at arbitrary, algorithm-chosen points — and its guarantee concerns the *global* coefficients $\hat{f}(S)$. For the *dataset* coefficients, plain samples are insufficient (Lemma 2.2); the main theorem (Theorem 2.2) restores power through CSAMP alone, with complexity governed by how much the dataset's contexts *repeat*, while the companion routes the dataset's structure through its *bias spectrum* (Lemma 2.5) at the cost of model queries.

2.2 Recap: the Classical Goldreich-Levin Algorithm

Theorem 2.1 (Goldreich-Levin [18]; [31] Section 3.5): Given query access to $f : \{-1,1\}^n \to \{-1,1\}$ and $0 < \tau \leq 1$, there is a poly$(n, 1/\tau)$-time algorithm that with high probability outputs a list $L = \{U_1, ..., U_\ell\}$ of subsets of $[n]$ such that:
1. $|\hat{f}(U)| \geq \tau \Longrightarrow U \in L$;
2. $U \in L \Longrightarrow |\hat{f}(U)| \geq \tau/2$.

By Parseval, the second guarantee implies $|L| \leq 4/\tau^2$.

It is a divide-and-conquer search over the tree of coordinate prefixes. For $S \subseteq J \subseteq [n]$, define the **bucket weight** ([31], Definition 3.39)

$$W^{S \mid J}[f] = \sum_{T \subseteq \overline{J}} \hat{f}(S \cup T)^2,$$

the Fourier weight of f on sets whose restriction to J equals S. The algorithm keeps the buckets of weight at least $\tau^2/2$ — at most $2/\tau^2$ of them by Parseval ($4/\tau^2$ once estimation slack is included) — splitting one coordinate at a time until each survivor is a singleton $\{S\}$ with $\hat{f}(S)^2 \geq \tau^2/2$. This is feasible because bucket weights are *estimable*: by the restriction identity ([31], equation (3.5) and Proposition 3.40),

$$W^{S \mid J}[f] = \mathbb{E}_{z \sim \{-1,1\}^{\overline{J}}}\left[\hat{f}_{J \mid z}(S)^2\right] = \mathbb{E}_{z \sim \{-1,1\}^{\overline{J}}}\ \mathbb{E}_{y, y' \sim \{-1,1\}^J}[f(y,z)\chi_S(y) \cdot f(y',z)\chi_S(y')],$$

so an algorithm with query access to f gets accuracy $\pm\epsilon$ with $O(\log(1/\delta)/\epsilon^2)$ queries.

Two facts for later. The proof only needs $\|f\|_\infty \leq 1$ (for the Chernoff estimates) and $\|f\|_2 \leq 1$ (for the Parseval pruning bound), so Theorem 2.1 holds verbatim for $f : \{-1,1\}^n \to [-1,1]$, with list size $4\|f\|_2^2/\tau^2$. And the restriction identity averages the *uniform* restriction z; the main theorem below replaces this uniform average by the dataset's own conditional distribution.

2.3 Obstructions

Why can't we just run the tree search on the dataset coefficients? The Mass Identity (Theorem 1.1) already warns us: $\sum_S \hat{f}_{\mathcal{D}}(S)^2 = C_{\mathcal{D}} \cdot \mathbb{E}_{\mathcal{D}}[f^2]$, so Parseval fails over $\mathcal{D}$ by the density factor $C_{\mathcal{D}}$. This kills the pruning bound — there can be up to $C_{\mathcal{D}}\mathbb{E}_{\mathcal{D}}[f^2]/\tau^2$ heavy coefficients, i.e. exponentially many — and not only for exotic f:

Example: Let $K \subseteq [n]$ and let $\mathcal{D} = \{x : x_i = 1 \text{ for all } i \in K\}$ be the subcube fixing K, with $f \equiv 1$. Every χ_S with $S \subseteq K$ is identically 1 on $\mathcal{D}$, so

$$\hat{f}_{\mathcal{D}}(S) = \begin{cases} 1 \text{ if } S \subseteq K \\ 0 \text{ otherwise,} \end{cases}$$

and there are $2^{|K|}$ maximal coefficients: distinct characters are *identical as functions on* $\mathcal{D}$, and "the" heavy set is only defined up to this aliasing.

Worse: the bucket weights themselves carry *no information* over a generic dataset. The following exact identity is the dataset analogue of the restriction identity.

Lemma 2.1 *(Pair Form of Dataset Bucket Weights)*: Fix $J \subseteq [n]$ with $|J| = k$ and $S \subseteq J$. Then

$$\sum_{U \subseteq \overline{J}} \hat{f}_{\mathcal{D}}(S \cup U)^2 = 2^{n-k} \cdot \mathbb{E}_{x,x' \sim \mathcal{D}}[f(x)f(x')\chi_S(x_J)\chi_S(x'_J) \cdot \mathbb{1}[x_{\overline{J}} = x'_{\overline{J}}]].$$

Proof: Expand each $\hat{f}_{\mathcal{D}}(S \cup U) = \mathbb{E}_{x \sim \mathcal{D}}[f(x)\chi_S(x_J)\chi_U(x_{\overline{J}})]$, square, and swap sums; the inner sum $\sum_{U \subseteq \overline{J}} \chi_U(x_{\overline{J}})\chi_U(x'_{\overline{J}})$ is the reproducing kernel $2^{n-k} \cdot \mathbb{1}[x_{\overline{J}} = x'_{\overline{J}}]$ on $\{-1,1\}^{\overline{J}}$. ■

Lemma 2.2 *(Blindness of Dataset Bucket Weights)*: In the setting of Lemma 2.1, suppose no two dataset points collide on $\overline{J}$ (the projection $x \mapsto x_{\overline{J}}$ is injective on $\mathcal{D}$). Then for *every* $S \subseteq J$,

$$\sum_{U \subseteq \overline{J}} \hat{f}_{\mathcal{D}}(S \cup U)^2 = \frac{2^{n-k}}{|\mathcal{D}|} \cdot \mathbb{E}_{x \sim \mathcal{D}}[f(x)^2],$$

independent of S.

Proof: Under injectivity only the diagonal of Lemma 2.1 survives, and $\chi_S(x_J)^2 = 1$. ■

In words: whenever the dataset is collision-free on the un-split coordinates, every level-k bucket has *identical* weight — regardless of f or where its heavy coefficients lie. As m random points collide on $\overline{J}$ only with probability $\binom{m}{2}/2^{n-k}$, the tree is information-free below the *birthday horizon* $k \approx n - 2\log_2 m$, so a GL search descends blindly through $2^{n-2\log_2 m}$ buckets before any weight tells buckets apart. With the subcube example, this rules out a sample-only dataset GL — extra access or structural repetition in $\mathcal{D}$ is necessary. It is the dataset-specific form of the statistical-query / noisy-parity barrier [6, 7, 28], and of the sample-only insufficiency that query-capable provers remove for learning [19, 22]; the missing power is restored below by context repetition (Theorem 2.2) or the bias spectrum (Theorem 2.3).[3]

2.4 Main Theorem: Goldreich-Levin from Context Conditioning

The blindness lemma tells us exactly what to look for: its collision-free hypothesis fails precisely when dataset points *share contexts*. Real corpora share contexts constantly — that is what n-gram statistics are — and CSAMP is the primitive that exposes the sharing. The headline algorithm replaces the classical bucket weight, whose restriction z is averaged over the *uniform* measure, by a conditional statistic whose contexts are drawn from *the dataset itself*.

Definition 2.2 *(Contexts and Context Groups)*: Fix $J = J_k = [k]$ and $\overline{J} = [n] \setminus J$. The *contexts* at level k are the distinct strings $Z_k = \{x_{\overline{J}} : x \in \mathcal{D}\}$ appearing on the un-split coordinates; the *context group* of $z \in Z_k$ is $\mathcal{D}_z = \{x \in \mathcal{D} : x_{\overline{J}} = z\}$ — the datapoints sharing that context. We write $z \sim \mathcal{D}_{\overline{J}}$ for a context drawn with the data ($z = x_{\overline{J}}$, $x \sim \mathcal{D}$), and $c_k = |Z_k|$ for the number of distinct contexts (the *context profile*).

[3]Sanity check for Lemma 2.2: take $n = 2$, $\mathcal{D} = \{(1,1), (-1,-1)\}$, $f(x) = x_1$, $J = \{1\}$. Then $\hat{f}_{\mathcal{D}}(\{1\}) = 1$, $\hat{f}_{\mathcal{D}}(\{1,2\}) = 0$, yet the bucket weight is $1 = \frac{2^{2-1}}{2} \cdot 1$ — signal and aliasing noise conspiring to a flat profile.

Definition 2.3 *(Conditional Bucket Weight)*: For $S \subseteq J$ and $z \in Z_k$, let $\overline{v}_S(z) = \mathbb{E}_{x\sim\mathcal{D}_z}[f(x)\chi_S(x_J)]$ — the "average with the string z fixed." The *conditional bucket weight* is

$$\Psi(S \mid J) = \mathbb{E}_{z\sim\mathcal{D}_{\overline{J}}}[\overline{v}_S(z)^2].$$

Lemma 2.3 *(Properties of the Conditional Bucket Weight)*: For all $S \subseteq J = J_k$:
1. (Tower) $\hat{f}_{\mathcal{D}}(S \cup U) = \mathbb{E}_{z\sim\mathcal{D}_{\overline{J}}}[\chi_U(z)\overline{v}_S(z)]$ for every $U \subseteq \overline{J}$;
2. (Completeness) $\Psi(S \mid J) \geq \hat{f}_{\mathcal{D}}(S \cup U)^2$ for every $U \subseteq \overline{J}$;
3. (Termination) $\Psi(S \mid [n]) = \hat{f}_{\mathcal{D}}(S)^2$;
4. (Monotonicity) $\Psi(S' \mid J_{k+1}) \leq \Psi(S \mid J_k)$ for both children $S' \in \{S, S \cup \{k+1\}\}$;
5. (Level mass) $\sum_{S\subseteq J} \Psi(S \mid J) = 2^k \cdot \mathbb{E}_{z\sim\mathcal{D}_{\overline{J}}}\left[\frac{\mathbb{E}_{x\sim\mathcal{D}_z}[f(x)^2]}{|\mathcal{D}_z|}\right] =: R_k \leq 2^k \cdot \|f\|_\infty^2 \cdot \frac{c_k}{|\mathcal{D}|}$.

Proof: Throughout, $\mathcal{D}_z$ is the context group of z (Definition 2.2) and $\overline{v}_S(z) = \mathbb{E}_{x\sim\mathcal{D}_z}[f(x)\chi_S(x_J)]$ is the average of $f\chi_S$ over the datapoints sharing context z.

(1) Tower. Group the average defining $\hat{f}_{\mathcal{D}}(S \cup U)$ by context. Once the context $z = x_{\overline{J}}$ is fixed, the factor $\chi_U(x_{\overline{J}}) = \chi_U(z)$ is constant across the group and pulls out of the inner average:

$$\hat{f}_{\mathcal{D}}(S \cup U) = \mathbb{E}_{x\sim\mathcal{D}}[f(x)\chi_S(x_J)\chi_U(x_{\overline{J}})] = \mathbb{E}_{z\sim\mathcal{D}_{\overline{J}}}\left[\chi_U(z)\underbrace{\mathbb{E}_{x\sim\mathcal{D}_z}[f(x)\chi_S(x_J)]}_{\overline{v}_S(z)}\right].$$

This is the law of total expectation, splitting the data by context.

(2) Completeness. Read (1) as the inner product of $\overline{v}_S$ with the unit vector χ_U (unit because $\chi_U(z)^2 = 1$): one inner product cannot exceed the norm of $\overline{v}_S$. Concretely, Cauchy-Schwarz on (1) gives

$$\hat{f}_{\mathcal{D}}(S \cup U)^2 \leq \mathbb{E}_z[\chi_U(z)^2] \cdot \mathbb{E}_z[\overline{v}_S(z)^2] = \Psi(S \mid J).$$

So a descendant coefficient is dominated by its ancestor bucket's weight — a heavy leaf forces every ancestor heavy, which is exactly what lets the search prune without discarding a heavy coefficient.

(3) Termination. At $J = [n]$ the un-split part $\overline{J}$ is empty, so there is one (empty) context whose group is all of $\mathcal{D}$; hence $\overline{v}_S = \mathbb{E}_{\mathcal{D}}[f\chi_S] = \hat{f}_{\mathcal{D}}(S)$ and $\Psi(S \mid [n]) = \hat{f}_{\mathcal{D}}(S)^2$. At the bottom of the tree the bucket weight is exactly the squared coefficient we are hunting for.

(4) Monotonicity. Descending one level moves coordinate $k+1$ out of the context and into J, so the level-$(k+1)$ context fixes *one fewer* coordinate: it is a coarsening of the level-k context, and each level-$(k+1)$ group is a union of level-k groups. Averaging over the coarser groups before squaring can only shrink the mean square (Jensen). For the child $S' = S$: the level-$(k+1)$ value is the average of the level-k values over the groups merged into z', i.e. $\overline{v}_S(z') = \mathbb{E}[\overline{v}_S(z) \mid z']$, so

$$\Psi(S \mid J_{k+1}) = \mathbb{E}_{z'}\left[\mathbb{E}[\overline{v}_S(z) \mid z']^2\right] \leq \mathbb{E}_{z'}\mathbb{E}[\overline{v}_S(z)^2 \mid z'] = \Psi(S \mid J_k).$$

For the child $S' = S \cup \{k+1\}$: put $w = f \cdot \chi_{S'}\left(x_{J_{k+1}}\right) = f \cdot \chi_S(x_J) \cdot x_{k+1}$. The level-$k$ context already fixes $x_{k+1} = \pm 1$, so on each level-k group $\mathbb{E}[w \mid z] = \pm\overline{v}_S(z)$, giving $\mathbb{E}_z[\mathbb{E}[w \mid z]^2] = \mathbb{E}_z[\overline{v}_S(z)^2] = \Psi(S \mid J_k)$; coarsening to level $k+1$ is the same Jensen step. Either way a child is never heavier than its parent, so once a bucket is pruned its descendants stay pruned.

(5) Level mass. Fix a context z. The points of its group $\mathcal{D}_z$ agree on $\overline{J}$, hence *differ* on J: the map $x \mapsto x_J$ is injective on $\mathcal{D}_z$, so $\{x_J : x \in \mathcal{D}_z\}$ is itself a dataset inside $\{-1,1\}^J$, on which $\overline{v}_S(z)$ is precisely the dataset coefficient at S. The Mass Identity (Theorem 1.1) then applies verbatim to this small dataset — its density constant is $2^k / |\mathcal{D}_z|$ — and yields

$$\sum_{S \subseteq J} \overline{v}_S(z)^2 = \frac{2^k}{|\mathcal{D}_z|} \mathbb{E}_{x \sim \mathcal{D}_z}[f(x)^2].$$

Taking $\mathbb{E}_{z \sim \mathcal{D}_{\overline{J}}}$ of both sides gives the stated identity. For the bound, $\mathbb{E}_{x \sim \mathcal{D}_z}[f^2] \leq \|f\|_\infty^2$ and

$$\mathbb{E}_{z \sim \mathcal{D}_{\overline{J}}}[1/ |\mathcal{D}_z|] = \sum_{z \in Z_k} \frac{|\mathcal{D}_z|}{|\mathcal{D}|} \cdot \frac{1}{|\mathcal{D}_z|} = \frac{c_k}{|\mathcal{D}|}.$$

So the live weight at level k is governed by the number of distinct contexts c_k: when contexts repeat ($c_k \ll |\mathcal{D}|$) there is little mass to spread, and few buckets can be heavy.■

Everything a GL search needs is here: heavy descendants keep their ancestors heavy (2), the search terminates at exactly the dataset coefficient (3), pruned buckets stay pruned (4), and the number of live buckets is controlled by the *repetition structure* of the data (5) — R_k is small exactly when contexts repeat ($c_k \ll |\mathcal{D}|$ at levels where 2^k is large). Estimation is the dataset analogue of the classical restriction identity: draw a context *from the data* and two strings sharing it.

Lemma 2.4 *(Estimating Ψ)*:

$$\Psi(S \mid J) = \mathbb{E}_{z \sim \mathcal{D}_{\overline{J}}} \mathbb{E}_{x,x' \text{ i.i.d.} \sim \mathcal{D}_z}[f(x)\chi_S(x_J) \cdot f(x')\chi_S(x'_J)].$$

Each experiment costs one SAMP draw (yielding x and its context $z = x_{\overline{J}}$) and one CSAMP draw ($x' \sim \mathcal{D}_z$); the product lies in $[-1,1]$ when $\|f\|_\infty \leq 1$, so $O(\log(1/\delta)/\epsilon^2)$ experiments estimate $\Psi(S|J)$ to accuracy $\pm\epsilon$ with confidence $1-\delta$.

Proof: For fixed z, independence of x, x' gives $\mathbb{E}[f(x)\chi_S(x_J)f(x')\chi_S(x'_J)] = \overline{v}_S(z)^2$; average over z. A SAMP draw x has $x_{\overline{J}} \sim \mathcal{D}_{\overline{J}}$ and, conditioned on its context, $x \sim \mathcal{D}_z$; Hoeffding does the rest. ■

Definition 2.4 *(Live-Bucket Profile)*: For a threshold τ, the *live-bucket profile* of $(\mathcal{D}, f)$ is

$$N_k = |\{S \subseteq J_k : \Psi(S \mid J_k) \geq \tau^2/4\}|, \qquad N = \max_{0 \leq k \leq n} N_k.$$

By Markov on the level mass (property 5) and monotonicity (property 4), $N_k \leq \min(4R_k/\tau^2,\ 2N_{k-1})$, the latter for $k \geq 1$; the instances after the theorem show how the repetition structure of $\mathcal{D}$ controls N.

Theorem 2.2 (Dataset Goldreich-Levin from Context Conditioning): Given SAMP and CSAMP access to $(\mathcal{D}, f)$ with $\|f\|_\infty \leq 1$, and $0 < \tau \leq 1$, $\delta \in (0, 1)$, there is an algorithm that with probability at least $1 - \delta$ outputs a list L satisfying:
1. (Completeness) $|\hat{f}_{\mathcal{D}}(S)| \geq \tau \Longrightarrow S \in L$;
2. (Soundness) $S \in L \Longrightarrow |\hat{f}_{\mathcal{D}}(S)| \geq \tau/2$;
3. (Complexity) on the same event, at most $O(nN \cdot \log(nN/\delta)/\tau^4)$ sampling experiments and $\mathrm{poly}(n, N, 1/\tau, \log(1/\delta))$ time.

Moreover, given a target $\epsilon \in (0, 1]$ and a bound M on the *dataset spectral norm* $\left\|\hat{f}_{\mathcal{D}}\right\|_1 := \sum_{S \subseteq [n]} |\hat{f}_{\mathcal{D}}(S)| \leq M$, running at threshold $\tau = \min(1, \epsilon C_{\mathcal{D}}/(2M))$ and estimating each recovered coefficient to accuracy $\tau\sqrt{\epsilon/8}$, the algorithm also returns the sparse reconstruction

$$g = C_{\mathcal{D}}^{-1} \sum_{S \in L} \tilde{f}_{\mathcal{D}}(S) \chi_S \quad \text{with} \quad \mathbb{E}_{x \sim \mathcal{D}}[(f - g)^2] \leq \epsilon,$$

at the cost of $O(\log(|L|\ /\delta)/(\epsilon\tau^2))$ further SAMP draws. The algorithm needs no advance knowledge of N, never evaluates f outside $\mathcal{D}$, and never tests membership: every string it touches is a datapoint.

The algorithm is the classical tree search with Ψ in place of W:
1. Maintain a set of live buckets (S, J_k), starting from $(\emptyset, J_0)$.
2. At level k, split each live bucket on coordinate $k + 1$ and estimate $\Psi(S' \mid J_{k+1})$ for both children to accuracy $\tau^2/8$, giving the t-th estimate performed (in any fixed processing order) confidence $1 - \delta/(2t^2)$ via Lemma 2.4; keep the children with estimate at least $3\tau^2/8$. Note the confidence schedule references only the running index t, so the algorithm is well-defined with no advance knowledge of N.
3. Output the surviving singletons at level n as the list L.
4. *(Reconstruction.)* For each $S \in L$, estimate $\hat{f}_{\mathcal{D}}(S)$ from SAMP draws to accuracy $\tau\sqrt{\epsilon/8}$, and return $g = C_{\mathcal{D}}^{-1} \sum_{S \in L} \tilde{f}_{\mathcal{D}}(S) \chi_S$.

Proof: Condition on every performed estimate being $\tau^2/8$-accurate. The t-th estimate uses fresh samples, so it fails with probability at most $\delta/(2t^2)$ regardless of which buckets earlier estimates caused to be explored; summing over the (adaptively determined, possibly unbounded) sequence of estimates, the total failure probability is at most $\sum_{t \geq 1} \delta/(2t^2) \leq \delta$.

Completeness. If $|\hat{f}_{\mathcal{D}}(S)| \geq \tau$, then by completeness (2) of Lemma 2.3 every ancestor bucket of S has $\Psi \geq \tau^2$, so its estimate is at least $7\tau^2/8 \geq 3\tau^2/8$ and the ancestor survives at every level; hence $S \in L$.

Soundness. A surviving leaf has estimate at least $3\tau^2/8$, hence $\Psi(S \mid [n]) \geq \tau^2/4$; by termination (3), $\hat{f}_{\mathcal{D}}(S)^2 \geq \tau^2/4$.

Complexity. On the good event, a surviving bucket at level k has true $\Psi \geq \tau^2/4$, so there are at most N_k of them; the level-mass identity (5) gives $N_k \leq 4R_k/\tau^2$ (Markov), and monotonicity (4) gives $N_k \leq 2N_{k-1}$ (children of dead buckets are dead). Hence at most $2nN$ estimates are performed in total, and the t-th costs $O(\log(t/\delta)/\tau^4) = O(\log(nN/\delta)/\tau^4)$ experiments.

Reconstruction. Condition on the good event, so completeness holds. Let $g^{\uparrow} = \mathcal{D} \circ f$ be the lift, with $\widehat{g^{\uparrow}}(S) = C_{\mathcal{D}}^{-1} \hat{f}_{\mathcal{D}}(S)$ (Lemma 1.1) and $f = g^{\uparrow}$ on $\mathcal{D}$. Exactly as in Theorem 1.3 — drop the indicator and apply Parseval in $L^2(\mu)$:

$$\mathbb{E}_{\mathcal{D}}[(f-g)^2] = C_{\mathcal{D}} \cdot \mathbb{E}_{\mu}\left[\mathcal{D} \cdot (g^{\uparrow} - g)^2\right] \le C_{\mathcal{D}} \cdot \mathbb{E}_{\mu}\left[(g^{\uparrow} - g)^2\right] = C_{\mathcal{D}} \sum_{S} \left(\widehat{g^{\uparrow}}(S) - \hat{g}(S)\right)^2.$$

Since $\hat{g}(S) = C_{\mathcal{D}}^{-1} \tilde{f}_{\mathcal{D}}(S)$ for $S \in L$ and 0 otherwise, the two factors of $C_{\mathcal{D}}^{-1}$ cancel the $C_{\mathcal{D}}$:

$$\mathbb{E}_{\mathcal{D}}[(f-g)^2] \le \underbrace{C_{\mathcal{D}}^{-1} \sum_{S \notin L} \hat{f}_{\mathcal{D}}(S)^2}_{\text{tail}} + \underbrace{C_{\mathcal{D}}^{-1} \sum_{S \in L} \left(\hat{f}_{\mathcal{D}}(S) - \tilde{f}_{\mathcal{D}}(S)\right)^2}_{\text{estimation}}.$$

For the *tail*, completeness gives $|\hat{f}_{\mathcal{D}}(S)| < \tau$ for every $S \notin L$, so $\sum_{S \notin L} \hat{f}_{\mathcal{D}}(S)^2 \le \tau \sum_{S \notin L} |\hat{f}_{\mathcal{D}}(S)| \le \tau \left\| \hat{f}_{\mathcal{D}} \right\|_1 \le \tau M$; with $\tau \le \epsilon C_{\mathcal{D}}/(2M)$ this is $C_{\mathcal{D}}^{-1} \tau M \le \epsilon/2$. For the *estimation* term, soundness and the Mass Identity (Theorem 1.1) give $|L|\,(\tau/2)^2 \le \sum_S \hat{f}_{\mathcal{D}}(S)^2 = C_{\mathcal{D}} \cdot \mathbb{E}_{\mathcal{D}}[f^2] \le C_{\mathcal{D}}$, so $|L| \le 4C_{\mathcal{D}}/\tau^2$; estimating each $\hat{f}_{\mathcal{D}}(S)$, $S \in L$, to accuracy $\tau\sqrt{\epsilon/8}$ (the product $f\chi_S \in [-1, 1]$, so Hoeffding and a union bound over L cost $O(\log(|L|\,/\delta)/(\epsilon\tau^2))$ samples) makes it at most $C_{\mathcal{D}}^{-1} \cdot |L| \cdot \tau^2\epsilon/8 \le \epsilon/2$. Summing, $\mathbb{E}_{\mathcal{D}}[(f-g)^2] \le \epsilon$. ■

(For a deterministic time budget: cap the live buckets per level at $\overline{N}$, abort on overflow, and guess-and-double $\overline{N}$ with confidence budget halved per run; the run with $\overline{N} \ge N$ never aborts on the good event, at the cost of $O(\log N)$ restarts.)

Three instances of the parameter N, spanning the possible regimes (all verified numerically):

- **Dense datasets.** Always $c_k \le \min(|\mathcal{D}|, 2^{n-k})$, so $R_k \le \min(2^k, C_{\mathcal{D}})$ and $N \le 4C_{\mathcal{D}}/\tau^2$: for $C_{\mathcal{D}} = \mathrm{poly}(n)$ the search is polynomial outright — using strictly weaker access than a membership tester.
- **Subcube-structured data.** For the subcube example above (fixing K, $f \equiv 1$): $R_k = 2^{|K \cap J_k|} \le 2^{|K|}$, matching the true output size $2^{|K|}$ — the search is output-sensitive, even though the density constant $C_{\mathcal{D}} = 2^{|K|}$ is itself exponential in $|K|$.
- **Generic sparse data.** Singleton context groups (one datapoint per context) give $\overline{v}_S(z) = f(x)\chi_S(x_J)$, so $\Psi(S|J) = \mathbb{E}_{\mathcal{D}}[f^2]$ for *every* S — flat — and $c_k = |\mathcal{D}|$, $R_k = 2^k \cdot \mathbb{E}_{\mathcal{D}}[f^2]$: the parameter degrades to the trivial 2^k exactly when Lemma 2.2 bites, and CSAMP degenerates to SAMP. The theorem's power is context repetition, and the blindness lemma says this is not an artifact.

Two further remarks. First, nothing forces the split order $1, 2, ..., n$: any coordinate order works, and the profile R_k — hence the runtime — depends on it. For sequence data the natural choice is to keep *contiguous* contexts (so each context group is a set of strings sharing an $(n-k)$-gram), and choosing the order to maximize repetition is a purely combinatorial preprocessing question on $\mathcal{D}$. Second, applying Theorem 2.2 to the constant function $f \equiv 1$ finds all heavy *biases* $b_V = \mathbb{E}_{\mathcal{D}}[\chi_V]$ at threshold θ in time $\mathrm{poly}(n, N(\mathbb{1}), 1/\theta)$ — recovering, on-distribution, exactly the side information that the companion result below consumes.

2.5 Companion: Dataset GL from Model Queries

This subsection — and only this subsection, together with its corollaries below — assumes one additional oracle: **model queries** (QUERY), i.e., f evaluable at arbitrary $x \in \{-1, 1\}^n$, including out-of-distribution points. This is realistic when f is a cheaply queryable model, and it enables a different route: never condition, but instead read the dataset's effect on the spectrum through one structural object.

Definition 2.5 *(Bias Spectrum)*: The **bias spectrum** of $\mathcal{D}$ is the family

$$b_V = \mathbb{E}_{x\sim\mathcal{D}}[\chi_V(x)], \quad V \subseteq [n]$$

— the dataset Fourier coefficients of the constant function 1. Note $b_\emptyset = 1$ and $|b_V| \leq 1$. For $\theta \in (0, 1]$ — we never take θ larger — write $B_\theta = \{V : |b_V| \geq \theta\}$ for the **θ-heavy bias set**; since $b_\emptyset = 1$, always $\emptyset \in B_\theta$, so $|B_\theta| \geq 1$.

Lemma 2.5 *(Convolution Identity)*: For any $f : \{-1, 1\}^n \to \mathbb{R}$ and any $S \subseteq [n]$,

$$\hat{f}_{\mathcal{D}}(S) = \sum_{V\subseteq[n]} b_V \cdot \hat{f}(S\Delta V),$$

where Δ denotes symmetric difference.

Proof: Expand f in its global Fourier expansion inside the dataset expectation:

$$\hat{f}_{\mathcal{D}}(S) = \sum_T \hat{f}(T) \cdot \mathbb{E}_{x\sim\mathcal{D}}[\chi_T(x)\chi_S(x)] = \sum_T \hat{f}(T) \cdot b_{T\Delta S},$$

using $\chi_T\chi_S = \chi_{T\Delta S}$; substitute $V = T\Delta S$. ■

The bias spectrum is exactly the aliasing structure of the subcube example: there $b_V = \mathbb{1}[V \subseteq K]$, and the convolution smears the single global coefficient $\hat{f}(\emptyset) = 1$ onto all $2^{|K|}$ sets $S \subseteq K$. This coefficient aliasing is the same mechanism that subsampling induces on the Walsh-Hadamard spectrum, exploited constructively — with a *designed*, invertible subsampling pattern — in sparse-transform algorithms [30, 34]; the difference here is that the dataset, and hence its aliasing pattern, is given rather than chosen. The blindness of Lemma 2.2 is the statement that for a generic dataset the bias spectrum is an exponentially wide field of $\sim \pm 1/\sqrt{|\mathcal{D}|}$ noise, which buries any bucket-level signal. Conversely, whenever the bias spectrum is *dominated by few heavy terms*, the dataset spectrum of f is a controlled deformation of its global spectrum — which the following theorem exploits.

Theorem 2.3 (Dataset Goldreich-Levin, model-query access): Let $f : \{-1, 1\}^n \to [-1, 1]$ with $\|\hat{f}\|_1 \leq A$ be given by QUERY access, and let $\mathcal{D}$ be given by SAMP access. Let $0 < \tau \leq 1$, $\delta \in (0, 1)$, fix any $\theta \leq \tau/(4A)$, and suppose the θ-heavy bias set B_θ of $\mathcal{D}$ is given explicitly (e.g., via Theorem 2.2 applied to $f \equiv 1$, or from prior structural knowledge). Then there is an algorithm running in time $\mathrm{poly}(n, |B_\theta|, 1/\tau, \log(1/\delta))$, using $\mathrm{poly}(n, |B_\theta|, 1/\tau) \cdot \log(1/\delta)$ queries to f and $O(\log(|B_\theta| / (\tau\delta))/\tau^2)$ samples from $\mathcal{D}$, that with probability at least $1 - \delta$ outputs a list L satisfying:
1. (Completeness) $|\hat{f}_{\mathcal{D}}(S)| \geq \tau \Longrightarrow S \in L$;
2. (Soundness) $S \in L \Longrightarrow |\hat{f}_{\mathcal{D}}(S)| \geq \tau/2$;
3. $|L| \leq \frac{64\,|B_\theta|^3}{9\tau^2}$.

The algorithm:
1. Run classical GL (Theorem 2.1, in its $[-1, 1]$-valued form) on f with threshold $\tau' = \frac{3\tau}{4\,|B_\theta|}$ and confidence $1 - \delta/2$, obtaining a list F of global heavy sets, $|F| \leq 4/\tau'^2$.
2. Form the candidate list $C = \{T\Delta V : T \in F, V \in B_\theta\}$, so $|C| \leq |F| \cdot |B_\theta|$.
3. Draw $m = O(\log(|C| / \delta)/\tau^2)$ samples from $\mathcal{D}$ and, for each $S \in C$, compute the empirical coefficient $\tilde{f}_{\mathcal{D}}(S) = \frac{1}{m}\sum_{j=1}^m f(x^{(j)})\chi_S(x^{(j)})$. Output $L = \left\{S \in C : |\tilde{f}_{\mathcal{D}}(S)| \geq 3\tau/4\right\}$.

Proof: **Decomposition.** By Lemma 2.5, for every S,

$$\hat{f}_{\mathcal{D}}(S) = \sum_{V \in B_\theta} b_V \hat{f}(S \Delta V) + R_S, \quad \text{where } |R_S| \le \sum_{V \notin B_\theta} |b_V| \; |\hat{f}(S \Delta V)| \le \theta \cdot \left\|\hat{f}\right\|_1 \le \theta A \le \frac{\tau}{4},$$

using $|b_V| < \theta$ off the heavy set and re-indexing the tail sum over $T = S \Delta V$.

Completeness. Suppose $|\hat{f}_{\mathcal{D}}(S)| \ge \tau$. Then

$$\sum_{V \in B_\theta} |b_V| \; |\hat{f}(S \Delta V)| \ge |\hat{f}_{\mathcal{D}}(S)| - |R_S| \ge \tau - \frac{\tau}{4} = \frac{3\tau}{4},$$

and since $|b_V| \le 1$, some $V^* \in B_\theta$ has $|\hat{f}(S \Delta V^*)| \ge \frac{3\tau}{4\,|B_\theta|} = \tau'$. By the completeness of classical GL (except with probability $\delta/2$), $S \Delta V^* \in F$, hence $S = (S \Delta V^*) \Delta V^* \in C$. By Hoeffding and a union bound over C (except with probability $\delta/2$), every empirical coefficient satisfies $|\tilde{f}_{\mathcal{D}}(S) - \hat{f}_{\mathcal{D}}(S)| \le \tau/8$; in particular $|\tilde{f}_{\mathcal{D}}(S)| \ge \tau - \tau/8 > 3\tau/4$, so $S \in L$.

Soundness. If $S \in L$ then $|\hat{f}_{\mathcal{D}}(S)| \ge 3\tau/4 - \tau/8 = 5\tau/8 \ge \tau/2$.

List size and complexity. $|L| \le |C| \le |F| \; |B_\theta| \le \frac{4}{\tau'^2} \; |B_\theta| = \frac{64\,|B_\theta|^3}{9\tau^2}$. Classical GL at threshold τ' costs $\mathrm{poly}(n, 1/\tau') = \mathrm{poly}(n, |B_\theta|, 1/\tau)$ queries; step 3 costs m samples and $m \cdot |C|$ arithmetic operations. ∎

Note that the algorithm uses only the *set* B_θ, never the values b_V; and the theorem is meaningful precisely in the distribution-shift regime where $\hat{f}_{\mathcal{D}}(S)$ and $\hat{f}(S)$ differ — the shift is routed through the dataset's heavy biases. Running the completeness argument purely combinatorially (Parseval bounds the number of τ'-heavy global coefficients by $1/\tau'^2$) also yields an unconditional structural bound:

Corollary 2.1 *(List-Size Bound under Sparse Bias Spectrum)*: If $f : \{-1,1\}^n \to [-1,1]$ with $\left\|\hat{f}\right\|_1 \le A$, and $\theta \le \tau/(4A)$, then

$$|\left\{S : |\hat{f}_{\mathcal{D}}(S)| \ge \tau\right\}| \le \frac{16\,|B_\theta|^3}{9\tau^2}.$$

Representative Datasets

At the opposite extreme, if the dataset's spectrum uniformly tracks the global one — the "no distribution shift" regime — dataset GL degenerates gracefully to classical GL plus re-scoring.

Definition 2.6 *(ϵ-Representative Dataset)*: $\mathcal{D}$ is **ϵ-representative** for f if $|\hat{f}_{\mathcal{D}}(S) - \hat{f}(S)| \le \epsilon$ for all $S \subseteq [n]$.

Lemma 2.6: Let $f : \{-1,1\}^n \to [-1,1]$ and let $\mathcal{D}$ consist of m i.i.d. uniform samples. If $m \ge \frac{2}{\epsilon^2}\left(n \ln 2 + \ln \frac{2}{\delta}\right)$, then with probability at least $1 - \delta$, $\mathcal{D}$ is ϵ-representative for f.

Proof: For fixed S, $\hat{f}_{\mathcal{D}}(S)$ averages m i.i.d. terms $f(x)\chi_S(x) \in [-1,1]$ with mean $\hat{f}(S)$; Hoeffding gives $\Pr\left[\left|\hat{f}_{\mathcal{D}}(S) - \hat{f}(S)\right| > \epsilon\right] \le 2e^{-m\epsilon^2/2}$. Union bound over all 2^n sets. (Here $\mathcal{D}$ is the sampled *multiset* and $\hat{f}_{\mathcal{D}}(S)$ its multiset average; distinctness of points plays no role in this lemma or the proposition below.) ∎

Proposition 2.1 *(Dataset GL for Representative Datasets)*: Suppose $\mathcal{D}$ is $(\tau/8)$-representative for $f : \{-1,1\}^n \to [-1,1]$, and we have QUERY access to f and SAMP access to $\mathcal{D}$. Then the search problem of Definition 2.1 is solvable in time $\mathrm{poly}(n, 1/\tau, \log(1/\delta))$ with $O(\log(1/(\tau\delta))/\tau^2)$ samples: run classical GL at threshold $7\tau/8$

to get F, estimate $\hat{f}_{\mathcal{D}}(S)$ for $S \in F$ to accuracy $\tau/8$ from samples, and keep $|\tilde{f}_{\mathcal{D}}(S)| \geq 3\tau/4$.

Proof: If $|\hat{f}_{\mathcal{D}}(S)| \geq \tau$ then representativeness gives $|\hat{f}(S)| \geq \tau - \tau/8 = 7\tau/8$, so $S \in F$ by GL completeness; its estimate satisfies $|\tilde{f}_{\mathcal{D}}(S)| \geq |\hat{f}_{\mathcal{D}}(S)| - \tau/8 \geq 7\tau/8 > 3\tau/4$, so S is kept. (The representativeness slack is needed only for the GL step; the estimation step compares directly against $\hat{f}_{\mathcal{D}}(S)$.) If S is kept then $|\hat{f}_{\mathcal{D}}(S)| \geq 3\tau/4 - \tau/8 = 5\tau/8 \geq \tau/2$. GL's Parseval bound gives $|F| \leq 4/(7\tau/8)^2 \leq 7/\tau^2$, so a union bound over F needs $O(\log(|F|\ /\delta)/\tau^2)$ samples. ∎

In this regime the heavy dataset coefficients essentially *are* the heavy global coefficients; the substantive content of Theorem 2.2 and Theorem 2.3 lies exactly where representativeness fails.

2.6 Interaction Screening: Conditioning on the Interacting Set

Lemma 2.2 pinpoints why a sample-only dataset GL stalls: the context-conditioning estimator splits off J and conditions on the *complement* $\overline{J}$ of $n-k$ coordinates, whose subcube must repeat in $\mathcal{D}$. This fails above the birthday horizon $k \approx n - 2\log_2 m$: to reach a degree-d character the search needs $|J| \geq d$, i.e. $|\overline{J}| \leq n - d$, and the complement repeats only when $n - d \lesssim 2\log_2 m$ — a *short-context, high-degree* regime.

The complementary move conditions on the *interacting set itself.* For $S \subseteq [n]$ write $\nu_S := \mathbb{E}_z\left[(\mathbb{E}_{x\sim\mathcal{D}}[f(x) \mid x_S = z])^2\right]$, the energy of the best predictor using only the coordinates x_S. When $\mathcal{D}$ is a product distribution this is a bucket sum on the S-subcube,

$$\nu_S = \sum_{U\subseteq S} \hat{f}_{\mathcal{D}}(U)^2,$$

so the *pure* degree-$|S|$ interaction is recovered by Möbius inversion over subsets,

$$\hat{f}_{\mathcal{D}}(S)^2 = \sum_{U\subseteq S} (-1)^{|S|\,-\,|U|}\nu_U.$$

eq. (0) is the dual of Lemma 2.1 — split on S and keep the diagonal on S rather than on $\overline{J}$ — and is estimated by the U-statistic over dataset points sharing x_S,

$$\hat{\nu}_S = \frac{1}{m}\sum_g \frac{S_g^2 - Q_g}{n_g - 1}, \quad S_g = \sum_{x\in g} f(x), \quad Q_g = \sum_{x\in g} f(x)^2,$$

over the fibres $g = \{x \in \mathcal{D} : x_S = z\}$; singleton groups drop out, removing exactly the self-pair diagonal that Lemma 2.2 showed swamps the suffix version. This needs only the S-subcube — $2^{|S|}$ cells — to repeat, i.e. $|S| \lesssim \log_2 m$, *independent of* n: it reaches a sparse interaction inside an arbitrarily long context, at the price of a degree cap.

Proposition 2.2 *(Interaction Screening)*: Fix $d \leq \log_2 m - O(1)$ and $\tau > 0$. From SAMP access to $\mathcal{D}$ alone, conditioning on each S with $|S| \leq d$, the U-statistic estimator of eq. (0) together with eq. (0) recovers every S with $|S| \leq d$ and $\hat{f}_{\mathcal{D}}(S)^2 \geq \tau^2$, in time $O\big(\binom{n}{d} \cdot m \cdot 2^d\big)$ — with no repetition required on any complement $\overline{J}$, hence no birthday-horizon obstruction. It is *not* blind in the reverse direction: a pure degree-d character ($\hat{f}_{\mathcal{D}}(S) \neq 0$ but $\hat{f}_{\mathcal{D}}(U) = 0$ for all $U \subsetneq S$) has $\nu_S = \hat{f}_{\mathcal{D}}(S)^2$ and $\nu_U = 0$ for $U \subsetneq S$, so it is detected.

Proposition 2.2 is functional ANOVA — the Hoeffding–Sobol decomposition — read in the dataset-Fourier basis, and it is precisely what sample-only access *can* do: screening $\binom{n}{d}$ sets matches the statistical-query lower bound $n^{\Omega(d)}$ for recovering a size-d junta from random examples [7, 28], and no sample-only method does essentially better without breaking the noisy-parity assumption. The extra power the dataset model buys is the stronger *subcube-conditioning* access [9, 11] — which learns a d-junta in $2^{O(d)}$ rather than $n^{\Omega(d)}$ — realized here as "retrieve rows sharing this partial context," available exactly where $\mathcal{D}$ contains the conditioned subcube. The two primitives are complementary, tiling the (degree d, dimension n) plane by the single quantity $\min(d, n-d)$:

primitive	**conditions on**	**available when**
context-conditioning (Theorem 2.2)	complement $\overline{J}$, size $n-d$	$n-d \lesssim \log_2 m$ (high degree, short context)
interaction screening (Proposition 2.2)	set S, size d	$d \lesssim \log_2 m$ (low degree, long context)

The uncovered corner $\min(d, n-d) \gtrsim \log_2 m$ — high degree *and* long context — is exactly the sample-only, subcube-empty regime where Lemma 2.2 and the statistical-query / noisy-parity barrier [7, 28] leave no efficient method. The boundary is soft, not sharp: shrinking the conditional means of eq. (0) toward their lower-order parents (empirical Bayes) degrades $\hat{\nu}_S$ gracefully as the $2^{|S|}$ cells thin, giving a data-adaptive effective degree; and a kernel/embedding relaxation of the fibre ("rows with $x_S \approx z$") trades bias for reach — the exact-to-soft spectrum that connects this combinatorial recovery to representation learning, which passes the worst-case barrier only by assuming the smoothness a parity lacks.

> **CLAUDE:** Empirically (categorical Householder version, `interaction_predict.py`): on TinyStories next-*character* (dense $V = 27$, window 24) the recovered degree-3 interactions carry held-out predictability (collision energy positive out of sample) and lift top-1 over the degree-$\lesssim 2$ baseline; on next-*word* (sparser) the same screen overfits (held-out degree-3 energy < 0) — the $2^{|S|} \lesssim m$ boundary made visible. The genuinely high-degree win stays in the context-conditioning corner (Poelwijk, Theorem 2.2).

2.7 Application: Kushilevitz-Mansour over Datasets

Classical GL powers the Kushilevitz-Mansour learning algorithm [29] ([31], Theorems 3.37-3.38): any f with $\left\|\hat{f}\right\|_1 \leq s$ — in particular any decision tree of size s — is learnable from queries in time $\mathrm{poly}(n, s, 1/\epsilon)$. The spectral-norm hypothesis is exactly the A of Theorem 2.3 — the same L^1-geometry exploited by agnostic decision-tree learning [20] — so the on-distribution analogue of the coefficient-collection step comes for free:

> **Corollary 2.2** *(Heavy On-Dataset Coefficients of Decision Trees)*: Let $f : \{-1,1\}^n \to \{-1,1\}$ be computable by a decision tree of size s (so $\left\|\hat{f}\right\|_1 \leq s$), given by QUERY access, and let B_θ be the θ-heavy bias set of $\mathcal{D}$ for some $\theta \leq \tau/(4s)$. Then all S with $|\hat{f}_{\mathcal{D}}(S)| \geq \tau$ can be listed in time $\mathrm{poly}(n, s, |B_\theta|, 1/\tau, \log(1/\delta))$.

From listing to learning: Corollary 2.2 recovers the heavy coefficients, and the reconstruction clause of Theorem 2.2 turns them into an $\epsilon_{\mathcal{D}}$-close approximation. That clause needs nothing but the recovered heavy list and a bound on the dataset spectral norm $\left\|\hat{f}_{\mathcal{D}}\right\|_1$, so — although stated there for the context-conditioning search — the same scaled reconstruction $g = C_{\mathcal{D}}^{-1} \sum_{S \in L} \tilde{f}_{\mathcal{D}}(S)\chi_S$ applies verbatim to the model-query list L of Corollary 2.2. This is the honest dataset analogue of "coefficient collection plus Parseval": the aliasing that made a naive raw-coefficient g fail is absorbed by the single factor $C_{\mathcal{D}}^{-1}$ (Theorem 1.2), the same normalization that fixes the low-degree learner (Theorem 1.3). TODO: Bound the dataset spectral norm from the global one: $\left\|\hat{f}_{\mathcal{D}}\right\|_1 \leq \|b\|_1 \cdot \left\|\hat{f}\right\|_1$ by Lemma 2.5, so a low-L^1 bias spectrum together with small $\left\|\hat{f}\right\|_1 \leq s$ makes the reconstruction of Theorem 2.2 efficient in the model-query setting — genuine on-dataset *learning* of size-s decision trees, not just coefficient listing.

2.8 Discussion and Open Directions

The most closely related work is the Active Fourier Auditor [2], which likewise estimates spectral properties of a model over a data distribution; it orthonormalizes the parity basis to that distribution (following [25]) and reports a few scalar functionals (robustness, individual/group fairness), whereas our dataset Goldreich-Levin keeps the fixed characters over the empirical measure and recovers the heavy coefficients themselves. See Section 3 for the fuller landscape (non-uniform Fourier, the samples-versus-queries barrier, subsampling aliasing, and Fourier auditing of trained models).

TODO:

- Non-uniform (weighted) datasets: the identities above should extend with $p(x)$ in place of $1/|\mathcal{D}|$; the reproducing-kernel steps are unaffected, the mass identity picks up $\|p\|_2^2$-type factors.
- General product alphabets and orthonormal bases (Hermite, tokens): the pair-form and blindness lemmas need only the reproducing kernel; the convolution identity needs the group structure ($\chi_S \overline{\chi_T} = \chi_{S-T}$), so state it over $\mathbb{Z}_q^n$ or general finite abelian groups. For token alphabets, CSAMP is literally n-gram retrieval, and Theorem 2.2 applies as stated.
- Choosing the split order: the profile R_k depends on the coordinate order; find the order minimizing $\max_k R_k$ (a combinatorial problem on the dataset's context statistics — related to building a good trie).
- CSAMP from a flat subsample only: estimating Ψ without a corpus index requires context collisions *within* the subsample — a birthday/U-statistic analysis; quantify the subsample size needed as a function of the context-group sizes.
- Characterize which real datasets (images, token corpora) have small context profiles c_k along some order, and how N behaves empirically; relatedly, when is the heavy-bias set B_θ small and low-degree?
- Lower bound: turn Lemma 2.2 into a formal sample-complexity lower bound for Definition 2.1 in the SAMP-only model (two datasets, same visible statistics, different heavy sets), and extend it to CSAMP with all context groups singletons.

3 Related Work

Our starting point — the Fourier analysis of functions on $\{-1,1\}^n$ — is classical [31]. What changes is the *measure*: we replace the uniform distribution on the cube by the uniform distribution on a finite dataset $\mathcal{D}$, *keep* the parity characters χ_S, and ask what of the Fourier toolkit survives. This one choice — fixed basis, empirical measure — forces the non-orthogonality defect $C_{\mathcal{D}} = 2^n / |\mathcal{D}|$, the bias-spectrum convolution, and the samples-only "blindness" barrier. We organize prior work around it.

Fourier analysis off the uniform measure. The best-understood non-uniform setting is the p-biased / product-measure analysis ([31], Chapter 8; for learning, [14]), where the basis is rebuilt to remain orthonormal, so Parseval still holds. A recent line pushes this to genuinely non-product distributions by *orthonormalizing the basis to the distribution*: the discrete Fourier expansion of [25] for correlated variables, its use for learning DNF under a fixed distribution [24], and, in uncertainty quantification, the arbitrary polynomial chaos expansion built from the empirical moments of a raw dataset [32]. All of these *re-orthonormalize* and thereby avoid any $C_{\mathcal{D}}$-type defect; we instead retain the fixed characters and confront the non-orthogonality directly, which is what makes the dataset spectrum a *convolution* of the global spectrum rather than a change of basis.

Heavy-coefficient recovery and access models. We adapt the Goldreich–Levin algorithm [18] and its use in the Kushilevitz–Mansour learner [29], extended to general finite abelian groups by [3]; the agnostic-learning refinements of [20] use exactly the bounded spectral-norm ($\|\hat{f}\|_1 \leq A$) geometry we assume. All of these operate under the *uniform* measure with query access; our results are the empirical-measure incarnation, and the same $\|\hat{f}\|_1$ hypothesis is what routes the dataset shift through the bias spectrum.

Why samples alone are insufficient. That sample-only access cannot isolate a heavy parity is the statistical-query / noisy-parity barrier [6, 7, 28]. The same phenomenon reappears in verification: PAC-verification and the "interactive Goldreich–Levin" of [19] and [22] let a sample-only verifier recover heavy coefficients *only* with the help of a query-capable prover. Our blindness lemma is a dataset-specific version of this barrier (a birthday-horizon on collision-free projections). Where those works restore power through an interactive prover, we restore it either through *conditional (subcube) sampling* over the empirical dataset — the oracle studied for distribution testing and learning [9–11], here exposing the context repetition that real corpora exhibit — or, when the model is queryable, through the dataset's bias spectrum.

Aliasing as subsampling. The convolution identity $\hat{f}_{\mathcal{D}}(S) = \sum_V b_V \hat{f}(S \Delta V)$ is, mechanically, the aliasing that subsampling induces on the Walsh–Hadamard spectrum, exploited constructively in sparse-transform algorithms [30, 34] and in compressed-sensing recovery of sparse set functions [4, 36], including in explicitly non-orthogonal bases [37]. The distinction is that those methods *design* an invertible subsampling pattern with full query access, whereas our dataset is *given*: its induced aliasing (the bias spectrum) is not ours to choose, which is precisely why samples alone are blind and why the bias spectrum must be supplied.

Auditing and interpreting trained models via Fourier. Closest in motivation is the Active Fourier Auditor [2], which estimates robustness and (individual/group) fairness of a model from queries and samples, building on the fairness-verification line of [15–17] and the query-based auditing of [38]. That work also leaves the uniform measure — but by Gram–Schmidt-orthonormalizing the parity basis to the data distribution (the expansion of [25]) and estimating a handful of scalar functionals of the spectrum; we keep the fixed characters over the empirical measure and target the spectrum itself. A parallel, independent thread recovers the *sparse Walsh–Hadamard spectrum of a trained model* for explanation [21, 27], and — nearest to our machinery — [33] searches an *in-distribution* activation support for non-redundant heavy coefficients via an explicit "Goldreich–Levin with problem-specific constraints"; dataset Goldreich–Levin is the missing recovery primitive underneath these applications.

Sensitivity, learnability, and testing. Our per-coordinate sensitivity $\mathrm{Sens}^i_{\mathcal{D}}$ is the Fourier form of the total-effect Sobol index of variance-based global sensitivity analysis [35], whose subtlety under *dependent* (non-product) inputs [26] is the classical shadow of the $C_{\mathcal{D}}$ corrections; the same variance-based influence, estimated over a dataset, underlies feature-attribution methods such as [12]. The question of whether a model class can fit a given dataset is distribution-specific learnability: learnability under a *fixed* measure is governed by its metric entropy [5], and, spectrally, by how the target's weight aligns with the data-dependent spectrum [1, 8] — an information-theoretic cousin being dataset difficulty relative to a model family [13]. Finally, our in-distribution testing (which zeroes points outside $\mathcal{D}$) is close in spirit to distribution-free property testing [23], measuring distance over the data distribution rather than the uniform one.

4 Conclusion and Future Directions

Taking expectations over a dataset while keeping the characters of the ambient product space gives a sample-estimable Fourier calculus, provided one respects the density constant $C_{\mathcal{D}}$ and the aliasing it encodes. The dictionary (lift, inversion, mass identity, closeness) is elementary, and classical spectral algorithms transplant with it: low-degree learning survives with a scaled reconstruction, and Goldreich-Levin survives exactly when the dataset's bias spectrum is dominated by few heavy terms — with a matching impossibility (blindness) for sample-only access.

Beyond the open problems at the end of the Goldreich-Levin section, a few directions:

TODO:

- *Degree monotonicity.* Is there a monotone relationship between the degree (spectral concentration) of f over the dataset and over the full space? A clean statement here would let one read off, from dataset quantities alone, whether a given model class can learn f — connect to distribution-specific learnability under a fixed measure [5], to spectral notions of learnability via data-dependent eigenbases [1, 8], and to dataset difficulty relative to a model family [13].
- *The dataset as its own function.* What happens when $f = \mathcal{D}$ (the indicator itself)? The bias spectrum is then the whole story; relate its decay to standard notions of dataset complexity.
- *PAC learning and entropy.* Relate normalized weight profiles $\overline{W}^k_{\mathcal{D}}$ to PAC-learnability on-distribution, and to entropy-type quantities (an analogue of Hartley entropy for datasets).

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APPENDIX A Numerical Verification

Every identity, counterexample, and theorem constant in this note was verified numerically on random and structured instances, exact to 10^{-9}: the mass and closeness identities, the normalized Parseval identities, the pair-form and blindness lemmas, the convolution identity and its heavy-bias tail bound, all five properties of the conditional bucket weight Ψ together with the unbiasedness of its estimator, the subcube and half-cube aliasing examples, the dataset sensitivity bounds, the corrected learning bound, the $\epsilon_{\mathcal{D}}$-closeness reconstruction bound of the context-conditioning theorem (Theorem 2.2), and the candidate-generation constants of both Goldreich-Levin theorems. The self-contained script (Python, standard library only) lives at `verification/verify_identities.py` in the repository; run it with `python3 verification/verify_identities.py`.

TODO: Deferred proofs and the general finite-abelian-group statements, as they accumulate.

APPENDIX B Notes

CLAUDE: **Frame-theory reading (removed from the main text; kept here to revisit).** In frame-theory language: the mass identity (Theorem 1.1) says the restricted characters $\{\varphi_\alpha|_{\mathcal{D}}\}$ form a *tight frame* for $L^2(\mathcal{D})$ with frame bound $C_{\mathcal{D}} = |\mathcal{T}|^n \;/\; |\mathcal{D}|$; dividing each frame vector by $\sqrt{C_{\mathcal{D}}}$ — i.e., passing to the normalized characters of Definition 1.6 — yields a *Parseval frame*, for which every identity holds over the dataset with no constants at all. The normalized coefficients are the frame (analysis) coefficients, and the scaled reconstruction in Theorem 1.3 is the synthesis operator truncated to low degree. Aliasing is the frame's overcompleteness made visible: $|\mathcal{T}|^n$ frame vectors in a $|\mathcal{D}|$-dimensional space. Standard reference: Christensen, *An Introduction to Frames and Riesz Bases*.